

Implementation of a (4 × 4)–bit multiplier using registers and a down-counter

Objectives:

1. Use principles of repeated addition to implement a 4 x 4 multiplier.

Apparatus :

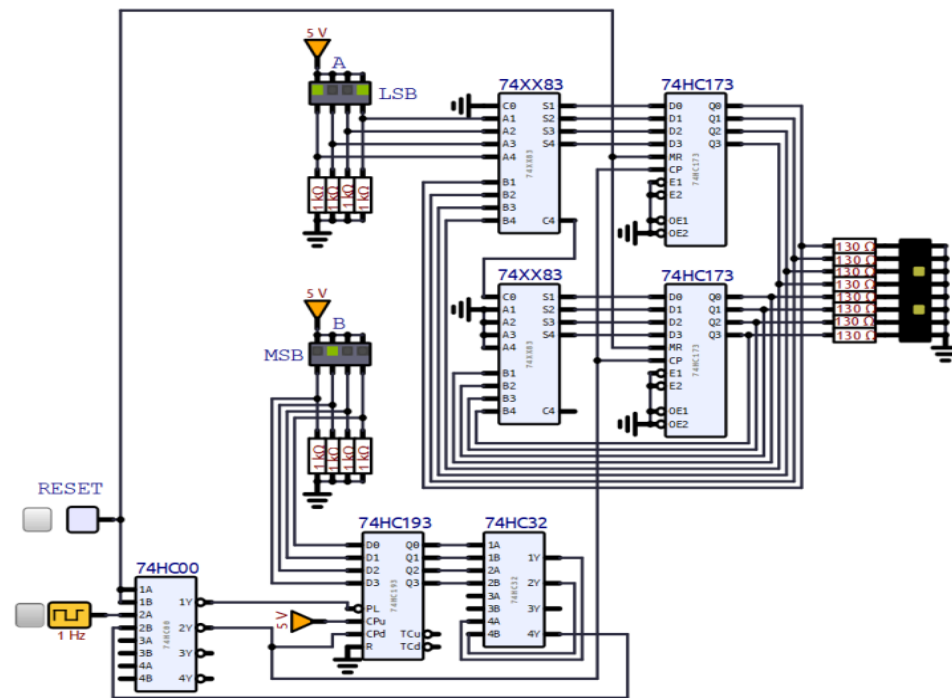
- IC-7483 – 4-bit adder (2)
- IC-74193 – Up/Down counter (1)
- IC-7432 – Quad OR gate (1)
- IC-74374 – 8-bit register (1)
- Resistors – 1 k Ω , (8)
- Breadboard
- DIP Switch – With 4 or more switches (2)
- Connecting wires
- DC Power Supply – 5 V (1)
- Function Generator (1)

Theory :

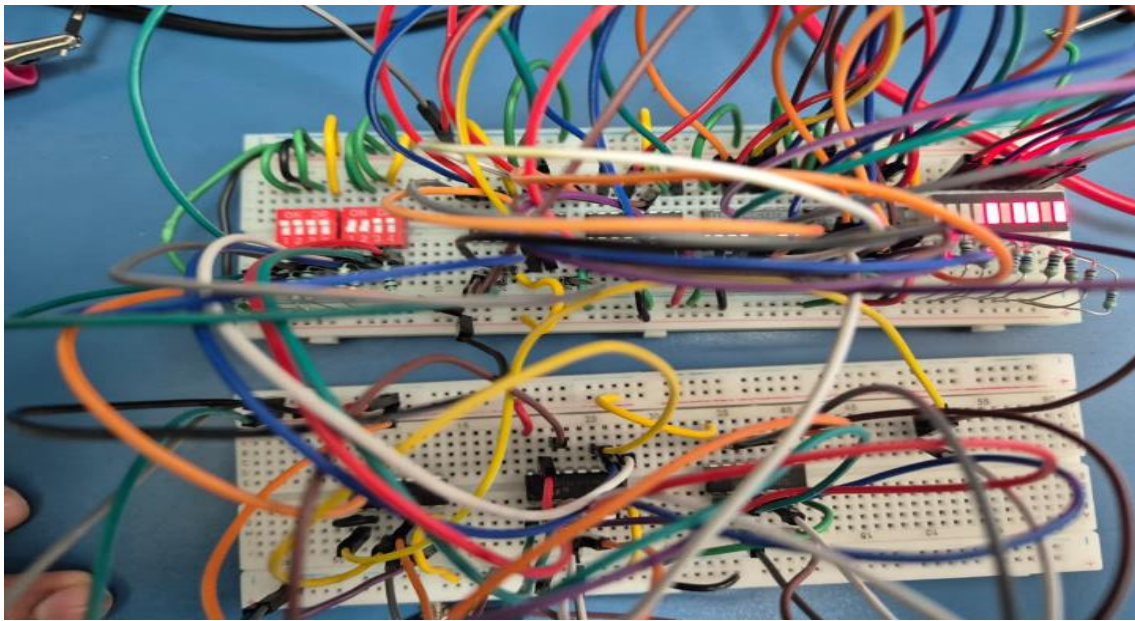
This project implements a **4 × 4-bit sequential binary multiplier** using the principle of **repeated addition**, one of the simplest hardware techniques for performing binary multiplication. Instead of generating the product in a single combinational step, the multiplicand is repeatedly added to an accumulator for a number of clock cycles equal to the value of the multiplier. This sequential approach reduces hardware complexity at the cost of increased computation time and forms the basis of many elementary arithmetic units used in digital systems.

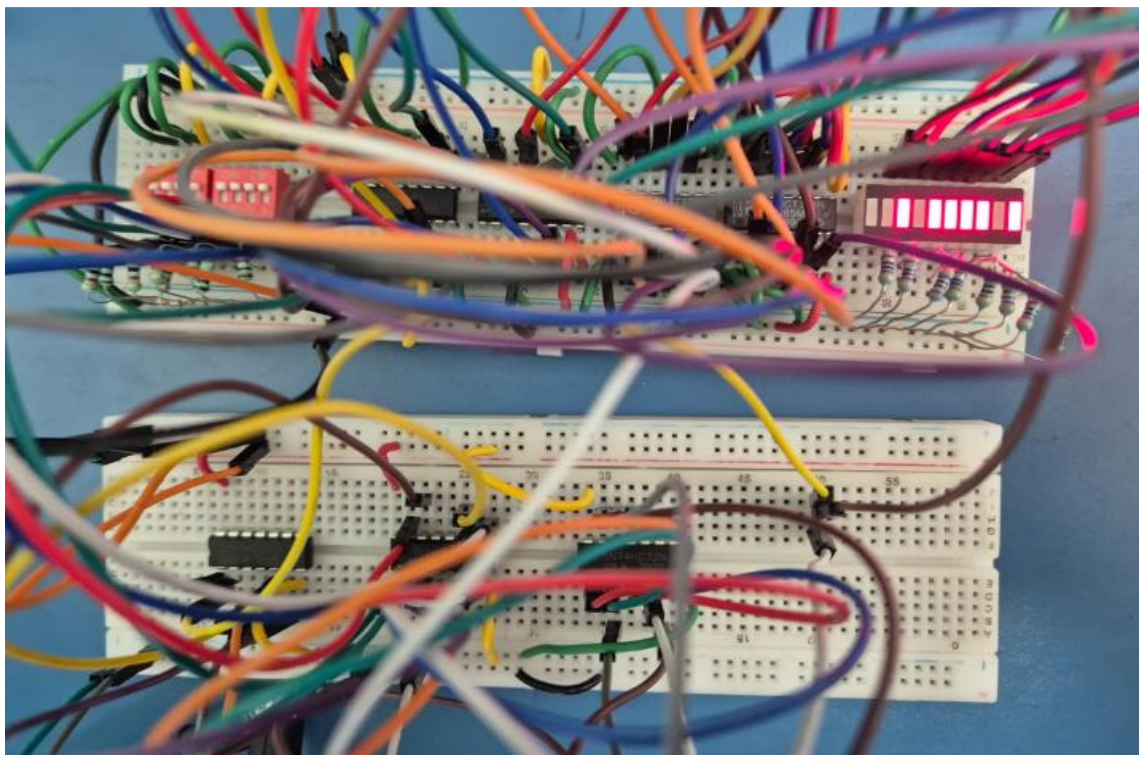
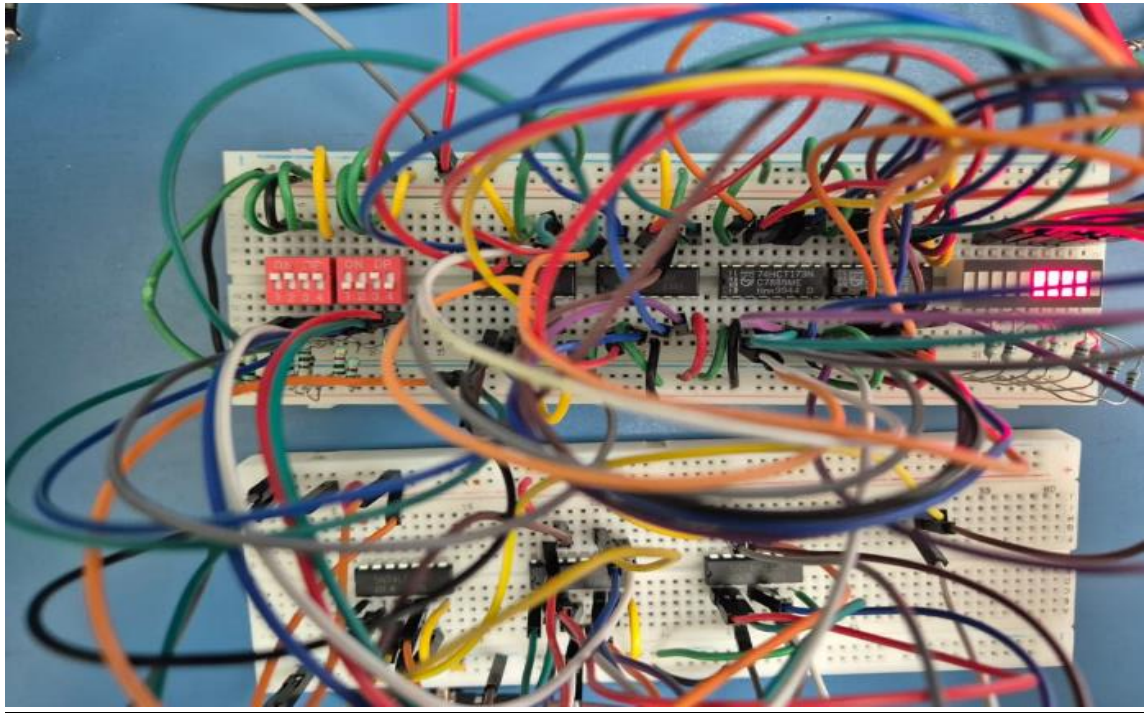
The design multiplies two **4-bit unsigned binary numbers** to produce an **8-bit product**. The hardware implementation consists of **two IC 7483 4-bit binary adders** cascaded to form an 8-bit adder, an **IC 74193 synchronous up/down counter** to count the remaining addition cycles, an **IC 74374 8-bit register** to store the accumulated product, and **IC 7432 OR gates** for the required control logic. A clock signal from a function generator synchronizes the sequential operation. During each clock cycle, the multiplicand is added to the current contents of the product register while the counter is decremented. The multiplication process terminates when the counter reaches zero, leaving the final product stored in the accumulator.

Circuit Diagram :



Observation :





Conclusion :

In this experiment, a 4×4 -bit binary multiplier was successfully implemented using digital logic components such as IC 7483 adders, IC 74193 down counter, IC 74374 register, and logic gates. The multiplication operation was performed using the principle of repeated addition, where the multiplicand was added repeatedly according to the value stored in the multiplier counter. The down counter controlled the number of additions, while the register stored intermediate and final results. With every clock pulse, the multiplicand was added to the product register until the counter reached zero, producing the final 8-bit product. This experiment demonstrated how arithmetic operations like multiplication can be implemented using sequential digital circuits and helped in understanding the working of adders, registers, counters, and control logic used in digital processors and arithmetic units.